

EXPERIMENT-5

AIM: To determine the viscosity of a given liquid using Ostwald's viscometer.

Requirements: Ostwald's viscometer, stop watch, specific gravity bottle, 10 cm³ pipette, distilled water, rubber tubing.

Principle: The Ostwald's Viscometer method is based on Poiseuille's equation. This relates the rate of flow of a liquid through a capillary tube with the coefficient of viscosity and is expressed by the following equation-

$$\eta = \pi r^4 t P / 8 V l \dots \dots \dots (1)$$

where,

V = Volume of the liquid of viscosity η flowing in time t, through a capillary tube of radius r and length l

P = Hydrostatic pressure of the liquid.

Thus, the determination of the absolute viscosity by means by Poiseuille's expression involves the determination of v, r, t, l and P. The method is however tedious and laborious one. Hence, a simpler method is used wherein we compare the viscosities of the two liquids. If the coefficient of viscosity of one liquid is known, then that of the other can be calculated.

If t_1 and t_2 are the flow times required to flow for equal volumes of two liquids through the same length of a capillary tube then from equation (1) we have

$$\eta_1 = \pi r^4 t_1 P_1 / 8 V l$$

$$\eta_2 = \pi r^4 t_2 P_2 / 8 V l$$

or

Where,

$$\eta_1 / \eta_2 = t_1 P_1 / t_2 P_2 \dots \dots \dots (2)$$

$$P = h d g \dots \dots \dots (3)$$

H = height of the liquid column (constant for all the liquids if taken for identical points and liquids are taken in equal volumes for a particular set of observations.)

g = acceleration due to gravity

d = density of the liquid

Since in this case for two liquids h and g are same hence,

$$\eta_1 / \eta_2 = t_1 P_1 / t_2 P_2$$

This will give us the relative viscosity of the given liquid. Relative viscosity has no units. The absolute viscosity

$$\eta_1 = d_1/d_2 \times t_1/t_2 \times \eta_2$$

Procedure:

1. Wash the viscometer with chromic acid ($K_2Cr_2O_7$ + Conc. H_2SO_4) and then rinse it 2-3 times with distilled water. It is finally washed with alcohol and ether and then dried.
2. Attach a piece of clean rubber tube to the end of viscometer and clamp the viscometer vertically in air.
3. Now introduce a sufficient volume of the given liquid with the help of a pipette in bulb so that the bend portion of the U-tube and more than half of bulb are filled up.
4. Through the rubber tube, suck up the liquid until it rises above the mark. Make sure that there is no air bubble inside the liquid.
5. Now, allow the liquid to fall freely through the capillary up to the mark. Start the stopwatch and note the time t_1 for the flow of the liquid from mark X to mark Y.
6. Repeat the experiment thrice. The values should be concordant.
7. Remove the liquid and clean and dry the viscometer again.
8. Repeat the experiment by taking the same volume of the distilled water and note down the time taken t_2 for flow of water from mark X to y Repeat thrice.
9. Weigh the relative density bottle and note down its weight.
10. Fill it with the given liquid and weigh it again.
11. Remove the liquid. Wash it with chromic acid and then distilled water. Dry in the oven. Now fill it with the distilled water and weigh it.

Observations:

Room temperature = $^{\circ}C$

Sr. No.	Given Liquid		Water	
	Time of Flow	Mean (t_1)	Time of Flow	Mean (t_2)
1.				
2.				
3.				

Weight of empty specific gravity bottle(W_1 g) = 18.008g

Weight of empty specific gravity bottle + given liquid(W_2 g)= 29.218g

Weight of empty specific gravity bottle + distilled water(W_3 g)=29.354g

Viscosity of Water = 18mp at 25 $^{\circ}C$

Calculation:

Density of solution/Density of water = $d_1/d_2 = w_2 - w_1 / w_3 - w_1$

Relative viscosity of the given liquid = $\eta_1/\eta_2 = d_1/d_2 \times t_1/t_2$

Absolute viscosity = $d_1/d_2 \times t_1/t_2 \times \eta_2$

Result:

The relative viscosity of the given solution isat $^{\circ}C$.

The absolute viscosity of the given solution is Pas at $^{\circ}C$.

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- Q1. What is the unit of coefficient of viscosity?
- Q2. Define viscosity of a liquid.
- Q3. How are the viscosity is measured and fluidity of a liquid related to each other?
- Q4. What is the effect of temperature on viscosity of liquid?